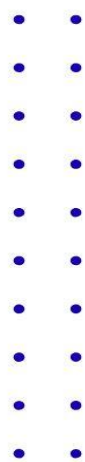


ARTIFICIAL INTELLIGENCE

DEEP DRIVE

AUTUMN 2026



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Total word count – 4357 words



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Preface

The DeepDrive company wants to convert any truck of them into self-driving autonomous truck. They want to show them at Consumer Electronics Show (CES) in a year. They already developed hardware of the truck but AI is still lagging. So, they want a detailed plan to develop the AI system.



Fig 1 – AI system in DeepDrive's Truck (magnific, 2026)

Task 1 – Overview of system

AI system for DeepDrive is complex. So, Different techniques and trade-off between them is necessary. The objective is to define plan with all technical information. In this section, overview of system is provided by comparing paradigm and methods. The goal is to build a demo of system in a year. So, cost and timeframe have to be clearly described.

a) Learning paradigms

Supervised learning

Supervised learning means training an AI by using labelled input-output data. Correct answers are used during training. For example, at DeepDrive, if they train AI by using thousands of road images labelled with correct steering angles, the AI in truck can analyze new camera image and produce correct steering action.

Advantages

Supervised learning is fast in training, easy to validation and strong in perception performance. It requires low computational cost.

Disadvantages

Supervised learning requires large number of labelled datasets. Annotation is expensive. It is limited in adaptability. It may struggle if it faces with unseen situations.

(Heintz, 2024)

Reinforcement learning

Reinforcement learning means training an AI through trial and error. AI gets rewards or penalties from its actions. For example, at DeepDrive, if they train an AI repeatedly driving in stimulated environment by getting rewards for safe actions and getting penalties for dangerous actions, the AI in truck can produce correct driving action in new situation.

Advantages

Reinforcement learning has strong sequential decision making, adaptable behavior and can optimize in long term driving objectives. It is useful in lane changing, speed control and platooning.

Disadvantages

Reinforcement learning needs high computational cost and long training time. Reward design is difficult and training is unstable. It is also unsafe for exploration that some actions may be dangerous. It is difficult to prove that system is safe in real roads as roads are unpredictable.

(Heintz, 2024)

Specific comparisons and trade-offs

Training time

Supervised learning only needs shorter time as correct targets are already known while reinforcement learning need longer time as AI need to be trained by many actions and episodes. So, Supervised is more suitable for DeepDrive truck project which has one year deadline.

Computational cost

In supervised learning, computational cost is moderate as it depends on dataset size and model complexity. In reinforcement learning, computational cost is high as it needs repeated simulations and policy updates. Then, reinforcement learning may produce advanced behavior but it requires more resources. The choice of learning model depends on the cost to spend.

Performance

Supervised learning is strong for perception tasks such as lane detection and traffic signs while reinforcement learning is strong for sequential control and decision optimization. Then, supervised learning can see environment effectively while reinforcement learning can decide how to act. So, for the truck in DeepDrive, both should be use to get the best performance.

Type of data

Supervised learning needs correct examples which are labelled by human. Reinforcement learning needs states, actions, rewards, penalties and interaction data which can get from stimulated environment. So, supervised learning needs high labelling costs while reinforcement learning needs high stimulation costs.

Explainability

Supervised learning is easier to explain as correct answers are already known. Reinforcement learning is harder to explain as it learns from rewards rather than labelled answers. So, for one year deadline, supervised learning is more suitable.

(Ilan Shaviv, 2026) (Specialist, 2026) (Rohan Inamdar, 2024)

b) Methods of learning

Neural networks

Neural networks are made by inspiring human brain which can learn patterns using layers of connected nodes. For example, at DeepDrive, they could use neural networks in trucks to recognize complex images, understand speech by natural language processing and do autonomous driving by making decisions from all inputs such as camera and sensor.

Advantages

Neural networks can learn complex patterns, discover useful features automatically and improve with new data. Then, they can be used in high complex non-linear problems. So, they are higher adaptability and scalability than expert systems and decision trees.

Disadvantages

Neural networks require larger data, higher computing power, much training time than expert systems and decision trees. They have high risk of overfitting when training for a long time with insufficient data and are also difficult to explain the reasons (blind-box decisions).

(Rawat, 2022)

Expert systems

Expert systems act as human experts by following predefined rules and facts. So, they are suitable for troubleshooting and rule-based systems. For example, at DeepDrive, they could use expert systems in trucks by giving clearly defined rules. So, truck can handle stably when facing with predefined conditions (e.g. it stops when facing with red light).

Advantages

Expert systems are predictable, auditable, consistent and does not need large training data but need reliable expert knowledge. They are more stable and better transparency(explainability) than neural networks and decision trees. They have less risks of overfitting.

Disadvantages

Expert systems cannot handle automatically when facing with new situations. They are higher maintenance, lower adaptability and scalability than expert systems and decision trees.

(Mohn, 2022)

Decision trees

Decision trees make decisions with sequence of conditions and branches (tree structure) and are suitable for simple to medium problems. For example, at DeepDrive, they could use decision trees in truck for risk assessment, classification and basic diagnosis about conditions.

Advantages

Decision trees are fast in training, easy to understand and needs fewer data than neural networks and expert systems.

Disadvantages

Decision trees are less accurate on complex tasks and can change significantly when data changes slightly. So, they cannot handle complex photos. They have also higher risk of overfitting than expert systems when the tree becomes too deep.

(Kavlakoglu, 2026)

Specific comparisons and trade-offs

Performance

Neural networks have highest performance for complex and unstructured tasks. When problem is predictable and completely covered by predefined rules, expert systems should be used as it has excellent performance among them in this condition. Decision trees have strong performance in simple and moderately complex problems. So, neural networks are most suitable for truck in DeepDrive company.

Possible pitfalls

Neural networks can have black-box decisions(unable to explain reasons), overfitting, biased outputs and high resource consumption. Expert systems can face rule explosion, outdated knowledge and fail when predefined rules do not match current new situations. Decision trees can have overfitting, reduced performance and instability when facing with high-dimensional problems.

Type of data

Neural networks can mainly handle unstructured data such as audio, video, images and text. Expert systems can handle structure facts, rules and expert knowledge. Decision trees can handle structured categorical and numerical data.

Explainability

Neural networks are hard to explain as internal reasoning is difficult to interpret(black-box decisions). Expert systems are very easy to explain as each decision follow predefined rules. Decision trees are easy to explain as decision path can be viewed as tree structure. So, expert systems are most suitable to use in truck of DeepDrive for explainability.

(geeksforgeeks, 2025) (indiafreenotes, 2026) (Donges, 2026)

Task 2 – Design of system

a) Appropriate method for each module

For controller module

Neural networks are suitable for controller module as they can handle large amount of data and different measurements although some measurements are inaccurate, missing or affected by interference. They can learn complex behaviors and produce continuous outputs.

For example, if they are used in truck of DeepDrive, they can collect large, different data and measurements from cameras, light detection and ranging (LiDAR) and radar. After data are collected, they learn about complex, non-linear driving behaviors which are difficult to explain. So, they can give continuous steering, braking and acceleration when driving on the road although facing with unpredictable situations. Neural networks are also widely used in modern autonomous driving systems.

Decision trees are not suitable for controller module as they will produce jerky control in steering, acceleration and braking when facing with hard decisions. If data are very complicated, they become difficult to maintain. Expert systems are not also suitable as they cannot learn and respond when facing with unfamiliar driving situations. They can happen rule explosion that human engineers would need to manually create rules for large number of road and traffic situations.

(milvus, 2026) (Roy, 2026)

For navigator module

Decision trees are suitable for navigator module as they provide fast processing, clear route selection and structured decision making by using tree structure. But they are suitable for simple to medium problems.

For example, if they are used in truck of DeepDrive, they can support structured decision makings by using clear conditions such as destination, traffic conditions, road conditions and online map data. So, truck can choose the roads based on distance, traffic, time and safety. As decision trees are fast to process and so they are suitable for real time route decisions and provide the most suitable navigation plan through a map.

Neural networks are not appropriate for navigator module as it is also limited transparency. Route planning needs clear and traceable decisions. Expert systems are also not suitable as they need human effort to maintain and can easily fail for large optimization tasks for navigating road systems.

(promwad, 2025) (Wang, 2025)

For safety module

Expert systems are suitable for safety module as they exactly follow predefined safety rules and explanation about decisions are easily to view. Verification about processes is also easier than black-box models.

For example, if they are used in truck of DeepDrive, they follow IF-THEN safety rules which are given by human experts based on obstacle distance, traffic, threat level and environmental conditions. So, truck can produce safety action such as emergency stop ON/OFF when detecting high threat level based on safety rules. It is also transparent and auditable and so human can check and easily verify for reason of emergency stop. Example rule, IF obstacle_distance < 2m THEN emergency_stop = ON.

Neural networks are not suitable for safety module as they are unpredictable and difficult to explain reason of action because of blind box decision. So, they are difficult for auditing and regulatory approval. Decision trees are not also suitable as they are less traceable and more difficult to update than expert systems.

(Abdallaoui, 2023) (rgbsi, 2026)

b) Controller module

Controller module consists of main vehicle control unit (VCU) which controls many electronic control units (ECUs). VCU which will be used in trucks of DeepDrive acts as vehicle's central brain which make decision. ECUs will be located in different locations in truck of DeepDrive such as input devices (camera, LiDAR, radar, GPS) and output devices (steer, throttle, brake, light).

So, VCU receives information from input devices, processes the data and sends commands to output devices that produce steering, braking, acceleration and warning systems.

(Adrian, 2025)

Controller inputs

Camera

Camera detects road scenes, lane markings, traffic signs, vehicles and pedestrians. It is excellent for object, lane detection and traffic sign recognition but its performance can decrease in fog, heavy rain, darkness or poor lighting. It captures image data or continuous video stream and sends them to VCU.

LiDAR

LiDAR uses lasers to measure distance to objects and creates three-dimensional shape. So, it can produce accurate 3D maps and detailed distance measurements to objects. But it is expensive and also affected by fog, rain and bad weather conditions. It uses continuous decimal distance values and sends them to VCU.

Radar

Radar uses radio waves to measure object distance, speed and movement. It is reliable in fog, rain, darkness and low visibility conditions. But it is lower resolution and less visual detail than cameras or LiDAR. It uses continuous numeric data for speed and distance and sends them to VCU.

GPS

GPS connects with satellites to produce vehicle position, coordinates and route location. It is essential for navigation, positioning and route planning. But signal can be lost in tunnels, near tall buildings or under dense cover. It uses continuous decimal latitude and longitude coordinates and sends them to VCU.

(Adrian, 2025)

Controller outputs

Steering

VCU can control steering for direction and position of the wheels. It uses angles as continuous decimal data such as -5° , 10° , or 25° to move steering for turning the truck smoothly.

Throttle

VCU can control throttle for engine or electric motor power. It uses continuous percentage or decimal data such as 30% or 75% to press throttle for decreasing or acceleration the truck.

Brake

VCU can control intensity of vehicle braking. It uses continuous percentage or decimal data such as 40% braking force to slow or stop the vehicle.

Light

VCU can control light as alerts to inform the driver. It uses discrete Boolean data such as ON or OFF to open or close the warning light.

(panasonic, 2023)

Complete example in truck of DeepDrive

Sense

When a truck is driving in motor way, controller inputs measure vehicle or surrounding environment. Then, they detect horse (obstacle) ahead. They send this information to the VCU.

Decide

The VCU receives information and processes it by comparing with safety limits. Then, the controller VCU determines that obstacle is very close.

Act

So, VCU commands ECUs which are located in controller output. ECUs control controller output (brake) to press the brake. Then, the vehicle slows down or stops to avoid a collision.

Feedback

Then, ECUs in controller inputs check that the vehicle is slowing sufficiently.

(Adrian, 2025) (panasonic, 2023)

c) Safety module

Safety module uses binary decision gate (1 or 0). For example, in truck of DeepDrive, when threat is detected, safety response is activated(1) and driving stops. When threat is not detected, it is not activated(0) and normal driving continues. It is used to protect truck, passengers and other road users by braking, slowing down or switching truck into safer lane.

It collects information from cameras, radar, LiDAR, GPS and other sensors. Then, it analyses data to identify pedestrians, obstacles, blocked lanes and other dangers. It compares detected situation with predefined safety rules and thresholds. It outputs 1 when danger is confirmed or 0 when danger is not detected. It activates braking or other safety action when output is 1.

(Drage, 2021)

False positive

False positive means system reports threat when real danger does not exist. For example, in truck of DeepDrive, when it is driving, harmless shadow or road sign is incorrectly identified as obstacle. So, it outputs 1 instead of outputting 0. Truck brakes or slows down unnecessarily. It reduces efficiency and makes users discomfort and delay time.

It is less dangerous than false negative although sudden braking can create risks. It makes system appear unpredictable, inefficient and disrupts surrounding traffic. It reduces trust of users that frequent unnecessary warnings can cause users to ignore alerts. So, false positive should be reduced without weakening hazard detection.

(Fussner, 2019)

False negative

False negative means system fails to detect real danger. For example, in truck of DeepDrive, when it is driving, real obstacle(pedestrian) is present but system does not activate brakes. It outputs 0 instead of outputting 1. Truck continues driving and causes collision, injury, death or system failure.

It is more dangerous than false positive as a real danger is not detected and it prevents safety system from performing its main purpose. It makes system unsafe and unsuitable for using. It reduces trust of users seriously as it can put users' lives in danger. So, it must be minimized as much as possible and is greater priority than false positive.

(Kosuru, 2022)

Sensitivity trade-off

Sensitivity	False negative	False positive
Increased	Reduce as more threats is detected.	Increase and cause unnecessary braking.
Decreased	Increase and miss real dangers.	Reduce as unnecessary interventions are reduced.

So, for trucks in DeepDrive, sensitivity should maintain balanced calibration to maintain safety while avoiding unnecessary interventions. But it needs extensive testing, validation and continuous monitoring.

(Britton, 2025)

Methods to minimize errors

For false positives, trucks in DeepDrive use better sensor fusion method which combines camera, radar, LiDAR and other data to create more accurate view. They use multiple-sensor confirmation that checks whether several sensors agree that threat exists. Threshold is also carefully tuning to prevent temporary noise, shadows or harmless objects from triggering system. They use context- aware detection that it detects not only object in isolation but also surrounding situations such as road and weather conditions.

For false negatives, trucks in DeepDrive use conservative safety thresholds that triggers more cautiously in high-risk situations. They use redundant sensors that provide backup detection if one sensor fails. Model is continuously retraining to improve recognition of new hazards and unusual situations. It is regularly testing with validation to identify weakness before truck is driven on public roads.

For both errors, real-time system monitoring (detects sensor faults, abnormal outputs) and fail-safe design (moves truck into safe state when system is malfunctioning) should be used.

(dataconomy, 2024)

Task 3 – Collection of data and training pipeline

a) Cost analysis with data collection phase

Data collection in practice and repetition

The sensor data are collected from cameras, radar, LiDAR and GPS through public road driving or simulations. Data collection must be repeated because the situations are always changing.

For DeepDrive, trucks must operate in different weather, traffic conditions and road types. So, one-time dataset cannot be used for every possible situation. Since DeepDrive is preparing reliable truck for CES demonstration, repeated data collection is necessary to test truck in changing environments. Repeated data collection can improve reliability and accuracy but it increases development time and operational costs.

(Hovhan, 2025)

Cost of collecting and labelling data

There is high cost for training which includes fuel, vehicle maintenance, specialist equipment, drivers, engineers and storage for large datasets.

The sensor data are recorded and manually annotated to important objects such as lanes, pedestrians, vehicles, traffic lights and road signs. Because supervised learning models require accurate labelled examples to learn correct relationship between collected sensor data and driving decisions. Labelling video and 3D LiDAR point clouds is also expensive and time consuming because it requires skilled annotators and specialist software.

For reinforcement learning, testing on real roads is expensive and not safe, so training should mainly use stimulations. Creating realistic simulations and repeated testing the AI can increase in computing (GPU training), reward engineering and development costs.

Since DeepDrive's truck development is progressing faster than AI software, high-quality labelled datasets are needed to produce quickly before CES. So, cost should be spent more in professional annotation to improve accuracy and safety because reducing it can make the prototype to perform poorly in complex road situations.

(Inamdar, 2024)

Data allocation for train, validation and test portions

Collected data is split into 70% for training data, 15% for validation data and 15% for test data. So, there is enough data for AI to validate and test but it uses less data for training than 80/10/10 splitting rule which can give more training data for neural network training and improve learning when total data is limited.

But 80/10/10 splitting rule has small validation and test sets. So, **70/15/15 splitting rule** is more appropriate for DeepDrive's truck because it balances AI learning with reliability, safety and accuracy when showing in CES.

(Gural, 2025)

b) Training, validation and testing phases

Training

Training cost and sources

Training AI of DeepDrive's truck requires large datasets, extensive simulations, storage and powerful computing hardware. Single computer is cheaper and easier to set up but it trains AI slowly. Multiple GPUs can be used for parallel processing to increase training speed. Cloud servers can also be used for powerful hardware access without major upfront costs. NVIDIA GPUs are commonly used for image processing and perception of AI training.

So, DeepDrive company should compare training time, hardware cost, cloud fees, electricity use and storage requirements when selecting the appropriate way. Faster AI training needs to increase hardware (multiple GPUs), power, cooling and data storage. It will be very expensive. If cloud servers are used in DeepDrive company for AI training, recurring fees and data transfer charges will be expensive too. So, cloud servers are more appropriate for DeepDrive because they cost less at start and can be used quickly to meet one year deadline for CES.

(Technology, 2023) (Fear, 2026) (team, 2025)

Time to stop training

Training should be stopped when additional training does not produce meaningful improvements.

For supervised learning, training should be stopped when AI's errors do not become smaller, accuracy stays the same after several trainings or validation performance begins to worsen. For reinforcement learning, training should be stopped when average reward remains stable in many trainings. Then, rewards should be monitored carefully.

So, the final AI should be tested that it performs well in new driving situations and has not only memorizes training environment. Training for a very long-time wastes resources such as computing, electric power, cooling and training time. It can also cause overfitting. But stopping early can result in undertrained AI model for trucks. So, DeepDrive company carefully manages the balance between overfitting and underfitting.

(Paniego, 2024)

Validation

Underfitting and overfitting

Validation can check that AI can generalize to new road situations rather than memorizing training data. Underfitting happens when the AI has not learnt enough from training data as training stops too early. So, it can fail to recognize different road signs, weather conditions or obstacles. Overfitting happens when AI learns training data including noise for much time and becomes performing poorly on unfamiliar roads (new situations).

So, DeepDrive company should monitor and compare training and validation performance. Poor results in both indicate underfitting. Strong training results but weak validation results indicate overfitting. There are also important risks such as biased datasets, poor quality or unbalanced sensor data and data leakage between training and validation data sets. So, DeepDrive company should also take care of these important risks to get the best AI for trucks in validation stage.

(Hodgson, 2022)

Testing

System evaluation in application

The autonomous truck of DeepDrive must be tested by using safety related behaviors. It should be tested in stimulation environment. Then, it should be moved to a closed test track for testing. After earlier testing stages, it can be tested in real-life environment by carefully supervised.

It should be tested with lane-keeping accuracy that measures lane deviation, collision rate that record collisions or near misses per kilometer, route completion that measures percentage of journeys completed successfully and emergency braking success that test however trucks stop safely before sudden hazards.

For testing the AI of truck in DeepDrive, the test data must remain completely unseen during development. Using test data during training or validation can produce unreliable results that can create serious safety risks.

(Aionlinecourse, 2025)

Task 4 – System deployment and safety

Safety must be one of the main factors for Autonomous trucks in DeepDrive company. So, possible failures and how to handle them must be clearly defined.

1. Heavy fog

Possible failure

Fog can scatter and absorb light which causes reducing image contrast and blurring boundaries. This can cause false negatives such as pedestrians being missed and false positives such fog being mistaken for an object. Detection accuracy can fall approximately from 91.5% (clear conditions) to 57.8% (heavy fog). So, this seriously reduces camera visibility. LiDAR can also become inaccurate because of signal scattering and noise.

Recovery strategy

The AI enters reduced-visibility mode which means reducing influence of camera data and focusing more on radar and LiDAR fusion. It reduces speed and increases following distance. So, the truck can stop within the distance at which cameras, radar or LiDAR can detect objects accurately.

Radar can perform well in this condition because microwaves penetrate fog and rain more effectively than visible light. Radar can provide reliable distance and relative-speed measurements while LiDAR can provide short-range 3D structural information. So, sensors should be used in combination as they have different failure characteristics.

(Liu, 2025)

2. Unexpected obstacle (fallen tree)

Possible failure

A fallen tree is a non-standard obstacle that may not appear in training data. It appears after a bend and extends across several lanes. The system could misclassify it as shadow or fail to recognize full size. This can cause serious collision as automatic swerving can damage other road users or excessive braking can cause wheel lock, skidding and loss of steering control.

Recovery strategy

The AI enters emergency obstacle response stage. It warns operator and nearby road users and then applies controlled emergency braking. After the car is stopped, the system enters safety mode. It secures vehicle, activates hazard lights and contacts control center.

Controlled braking reduces kinetic energy and is usually safer and more controllable than suddenly swerving. Radar can confirm that object is stationary while LiDAR estimates its dimensions and distance. Braking force can be adjusted for vehicle mass, road friction and available stopping distance.

(Bennett, 2021)

3. GPS failure

Possible failure

GPS spoofing and multipath errors are the most dangerous failures that system receives believable but incorrect position. The truck may believe that it is on another road. So, it can cause incorrect routing or unsafe maneuvers near junctions, bridges or restricted areas.

Recovery strategy

The AI enters navigation-degraded mode and then rejects GPS data which conflicts with other sensors. It uses IMU and wheel sensors to estimate truck's movement and position for a short time when GPS is unavailable. It matches visible features such as lane markings, signs and curbs against with HD local map. It maintains current lane and moves toward a safe stopping location when GPS is unavailable.

IMU measures acceleration, turning and orientation quickly while wheel sensors use wheel rotations to estimate speed and distance travelled. HD map matching uses stable environmental features and so the truck can maintain accurate position without depending entirely on satellite signals.

(Noh, 2024)

Conclusion

The DeepDrive company uses paradigms such as supervised learning and reinforcement learning and methods such as neural networks, expert systems and decision trees for training the AI system for their trucks.

It uses appropriate method for specific modules to train the AI system effectively. They analyze the cost of data collection in each training method. The AI system is trained by three phases such as training, validation and testing. They also consider possible failures and recovery strategies for safety. In this way, DeepDrive company add artificial intelligence system into their trucks which will show at Consumer Electronics Show in a year.

-----End of the Assignment-----

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